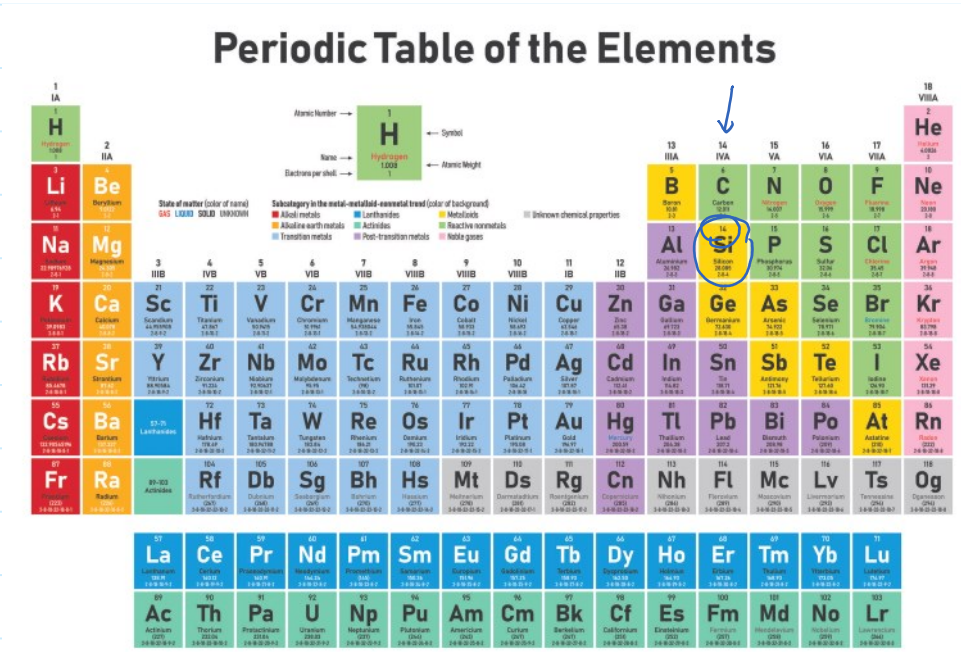
Reading

Adel S. Sedra and Kenneth C. Smith, **Microelectronic Circuits** 7th Edition, *Oxford University Press*, 2014.

- Chapter 3.1-3.3

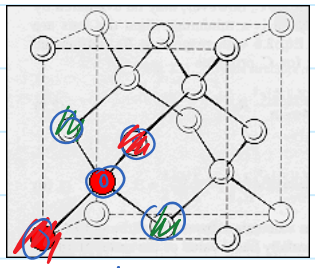


Si Atom

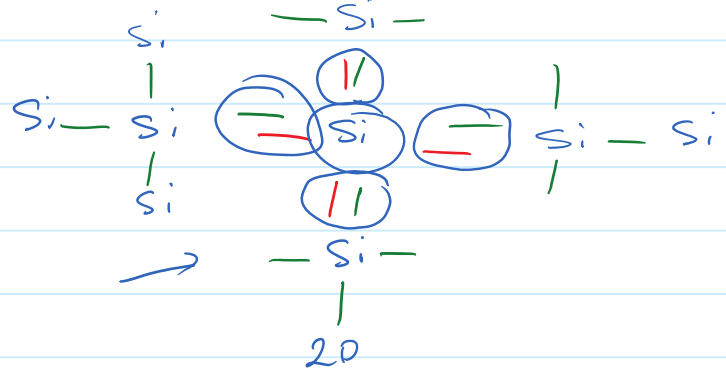


$14 \rightarrow 1s^2\ 2s^2 2p^6\ 3s^2\ 3p^2$ ✓ $3p^6$

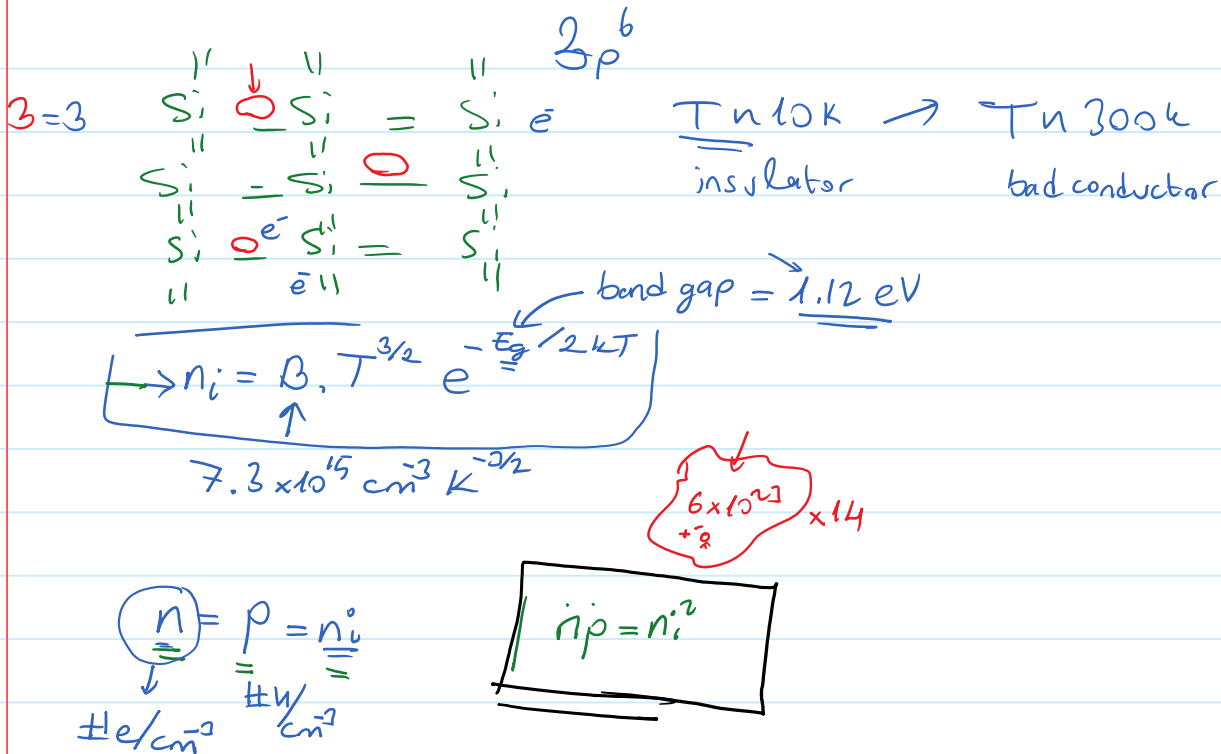
2 8 4



30

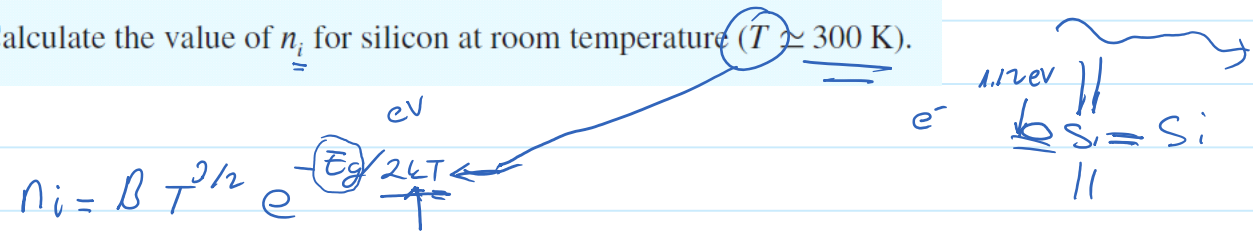

Si
Si
Si

Intrinsic Si



Example

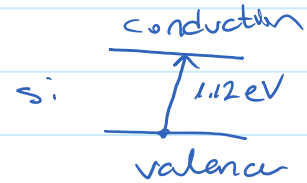
Calculate the value of n_i for silicon at room temperature ($T \simeq 300$ K).



$$n_i = B T^{3/2} e^{-E_g/2kT}$$

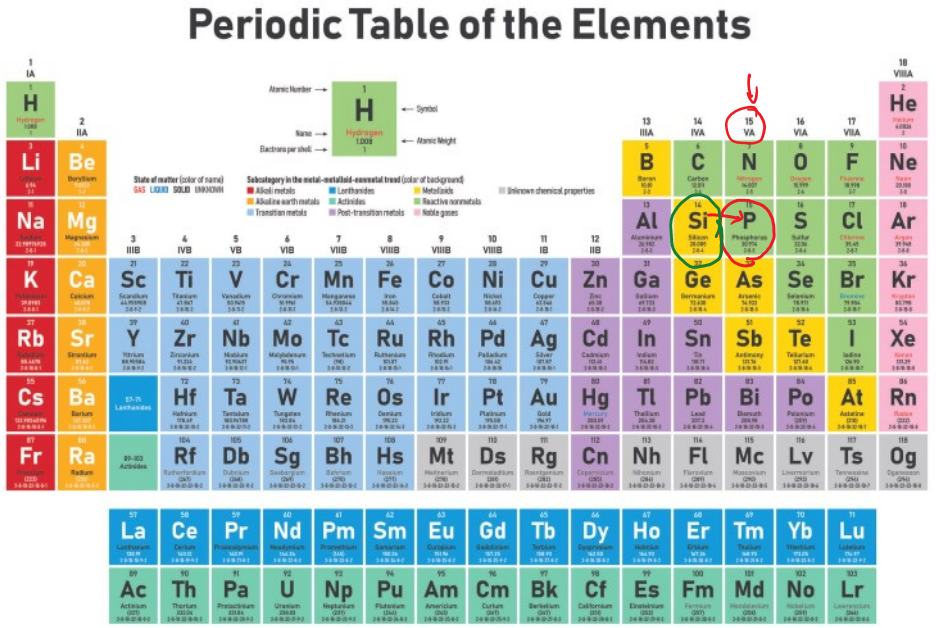
$$= 7.3 \times 10^5 \cdot 300^{3/2} \cdot e^{-1.12/(2 \times 8.62 \times 10^{-5} \times 300)}$$

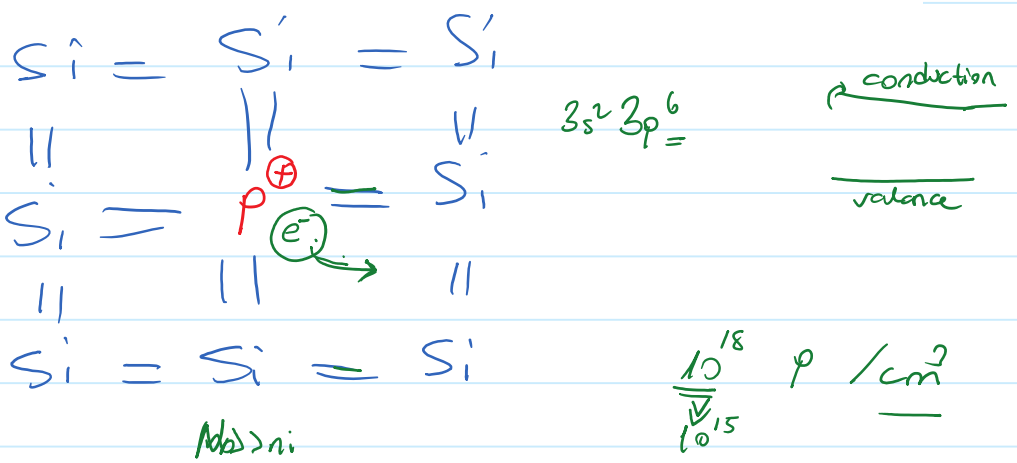
$$n_i = 1.5 \times 10^{10} \text{ carriers/cm}^3$$

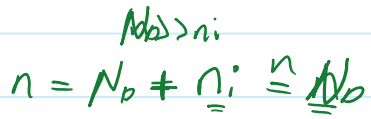


$\rightarrow n = n_i = p$

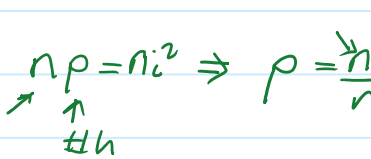
N-Type Semiconductor

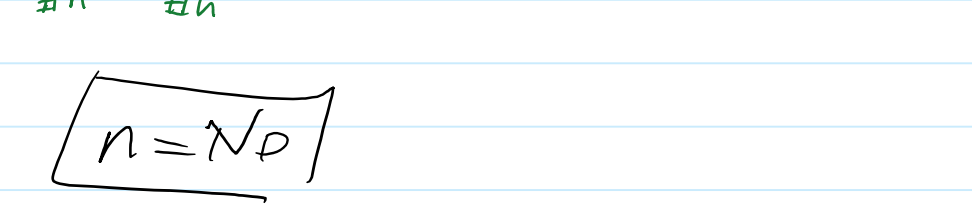




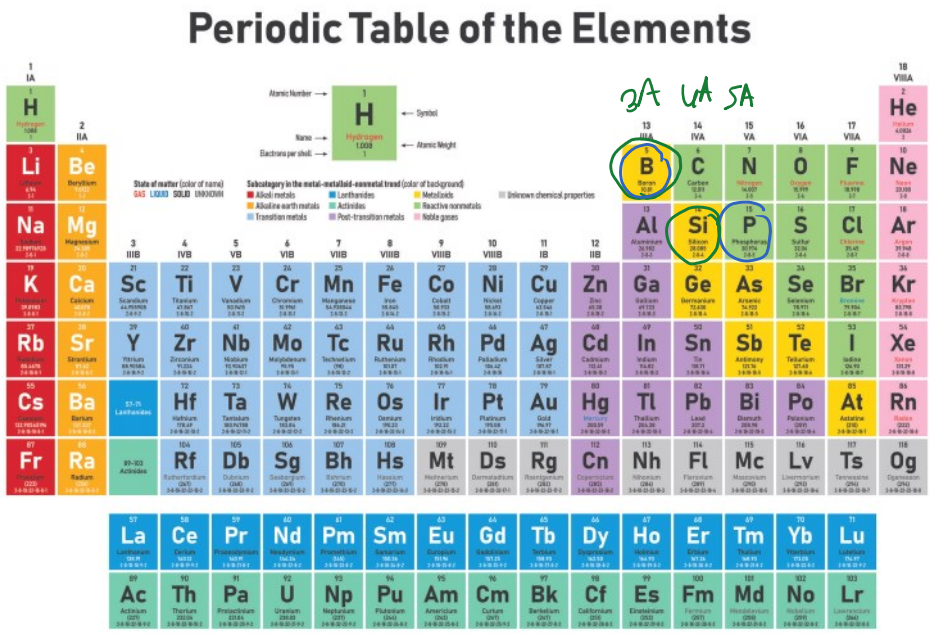








P-Type Semiconductor



$N_A \quad 10^{18} \; B \; cm^3$

$p = N_A$

p-type → hole → $p = N_A$

$np = n_i^2 \Rightarrow n = \frac{n_i^2}{p}$

1P → 1

$N_D \rightarrow \boxed{N_D = n}$

1B → 1h

$N_A \rightarrow N_A = p$

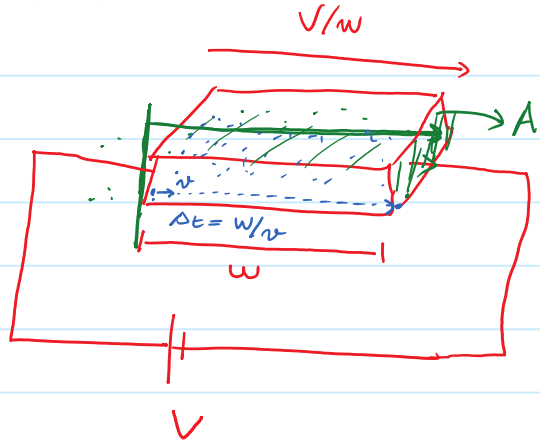
Example

Consider an n-type silicon for which the dopant concentration $N_D = 10^{17}/\text{cm}^3$. Find the electron and hole concentrations at $T = 300$ K.

$n = N_D = 10^{17}\ e^-/\text{cm}^3$

$p\,n = n_i^2 \Rightarrow p = \dfrac{n_i^2}{N_D} = \dfrac{(1.5 \times 10^{10})^2}{10^{17}} = 2.25 \times 10^3\ h/\text{cm}^3$

Drift Current



$v_p = E \cdot \mu_p$

$\frac{cm}{s}$ $\quad \frac{Holes}{cm^3}$

$\Delta Q = w \cdot A \cdot \rho \cdot q$

$I = \frac{\Delta Q}{\Delta t} = \frac{wA \cdot \rho \cdot q}{w/v}$

$I = \frac{wA\rho q}{w} \cdot E\mu_p$

$I/A = J = \rho q \cdot E\mu_p$

$J = \rho q E \mu_p - n q E \mu_n = E(q\rho\mu_p - qn\mu_n)$

σ

$\frac{1}{\sigma} = \rho$

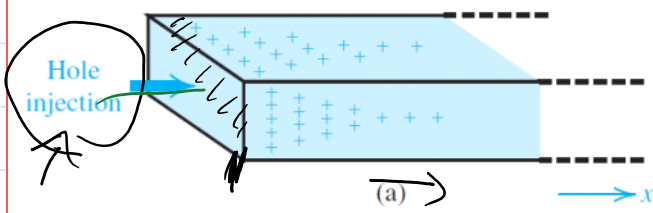
$R = \frac{\rho \cdot l}{A}$

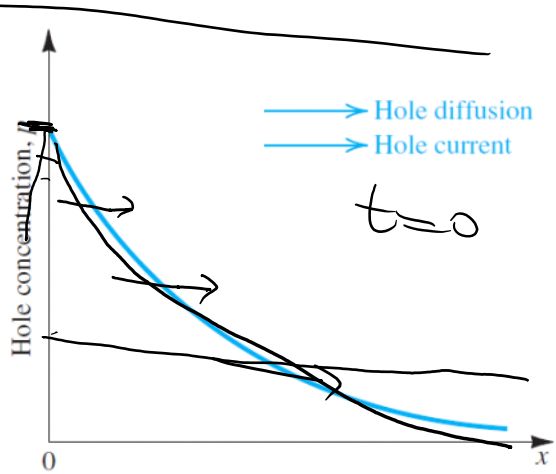
$J = \sigma E \Rightarrow \frac{I}{A} = \sigma \cdot \frac{V}{w} \Rightarrow V = \frac{1}{\sigma}\frac{w}{A} \cdot I$

Example

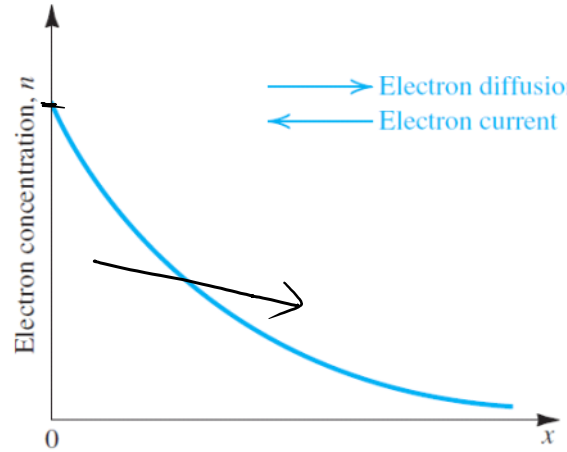
Find the resistivity of (a) intrinsic silicon and (b) p-type silicon with $N_A = 10^{16}/\text{cm}^3$. Use $n_i = 1.5 \times 10^{10}/\text{cm}^3$, and assume that for intrinsic silicon $\mu_n = 1350\ \text{cm}^2/\text{V} \cdot \text{s}$ and $\mu_p = 480\ \text{cm}^2/\text{V} \cdot \text{s}$, and for the doped silicon $\mu_n = 1110\ \text{cm}^2/\text{V} \cdot \text{s}$ and $\mu_p = 400\ \text{cm}^2/\text{V} \cdot \text{s}$. (Note that doping results in reduced carrier mobilities.)

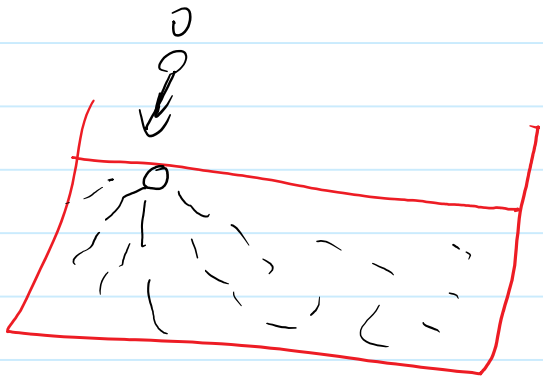
Diffusion Current

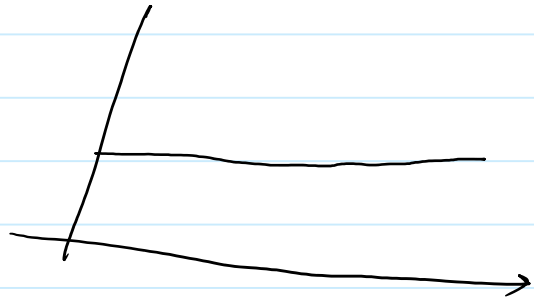




(b)





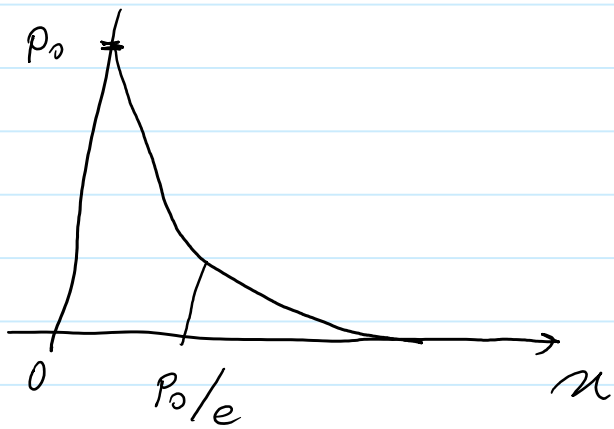


Example

Consider a bar of silicon in which a hole concentration profile described by

$$p(x) = p_0 e^{-x/L_p}$$

is established. Find the hole-current density at $x = 0$. Let $p_0 = 10^{16}/\text{cm}^3$, $L_p = 1\ \mu\text{m}$, and $D_p = 12\ \text{cm}^2/\text{s}$. If the cross-sectional area of the bar is $100\ \mu\text{m}^2$, find the current I_p.



$$J_p = -qD_p \frac{dp(x)}{dx}$$

$$J_n = qD_n \frac{dn(x)}{dx}$$

$$J_{p0} = q \frac{D_p}{L_p} p_0 = 192\ \text{A/cm}^2$$